

TOPIC 3.1

What are the properties of energy?

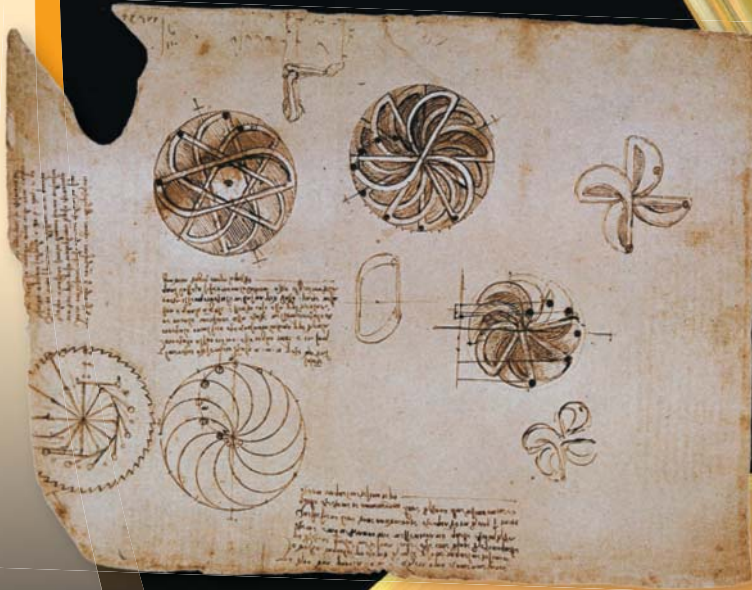
Key Concepts

- Energy can produce change in a system.
- There are different forms of energy.
- Energy can be transferred or transformed.
- Physical quantities contribute to different forms of energy.

Curricular Competencies

- Make observations aimed at identifying your own questions about the natural world.
- Use scientific knowledge to draw conclusions consistent with evidence.
- Generate and introduce new or refined ideas.

A perpetual motion machine is a machine that runs forever. For hundreds of years, it's been the dream of visionaries and gadget lovers alike. The famous Italian artist and thinker, Leonardo da Vinci, drew designs for several perpetual motion machines (inset photo). A modern reproduction of one of da Vinci's machines is shown in the main photo. A perpetual motion machine, were it to actually work, transfers and transforms energy, but energy is never lost. Once it is running, the machine does not need a push, fuel, or any other additional source of energy—ever. Despite his designs, da Vinci believed perpetual motion machines wouldn't, and couldn't, work. But that hasn't stopped countless people from trying to create one.





Starting Points

Choose one, some, or all of the following to start your exploration of this Topic.

1. **Reviewing** Think of what you already know about energy and its properties. Do you agree with da Vinci that a perpetual motion machine cannot exist? Why or why not?
2. **Modelling** Create a model of a perpetual motion machine. What forms of energy makes your machine work? How does your machine transfer or transform this energy as it runs?
3. **Applying First Peoples Perspectives** Most traditional First Peoples technologies and practices involve an understanding of how energy works. For example, using a deadfall trap or raising a totem pole requires knowledge of how energy can be used to move objects in particular ways and directions. Use a diagram to show how a traditional technology applies the use of energy.



Key Terms

There are five key terms that are highlighted in bold type in this Topic:

- system
- surroundings
- kinetic energy
- potential energy
- law of conservation of energy

Flip through the pages of this Topic to find these terms. Add them to your class Word Wall along with their meaning. Add other terms that you think are important and want to remember.

Energy can produce change in a system.

Activity

How Do You Describe Energy?



1. Think of an activity you are passionate about or interested in. What roles do energy play in it? For example, what forms does it take? What changes does it bring about? How does energy itself change? Record your ideas.
2. Discuss your ideas in small groups. Work together to come up with a description of what you think energy is.
3. Share and discuss your group's description with the class. Your teacher may add descriptions of energy from other sources as well.

Figure 3.1 Energy is all around you. **Analyzing:** How is energy present in this image?



What is **energy**? What does it do? How does it behave? Scientists have asked questions like these about energy for hundreds of years. Defining energy presents a problem, because it cannot be observed directly. Consider a scene like the one in **Figure 3.1**. Energy is present in many forms. It's in the air, a walk down the street, and even in a conversation. But it cannot be seen.

Despite this limitation, scientists can investigate energy indirectly. They do so by observing the effects it has on other things. Over time, by inquiring about such effects, scientists began to develop an understanding of the properties of energy. They found that

- energy can cause change in a system.
- there are different forms of energy, with different characteristics.
- these forms of energy can be transferred or transformed.
- different physical quantities contribute to different forms of energy.

Energy and Systems

system anything that is under observation

surroundings anything that is not part of a system

Anything that is under observation can be referred to as a **system**. For example, the person and the bungee cord in **Figure 3.2** could be considered a system. Everything that is not part of this system—that is, everything else in the entire universe—is considered the **surroundings**. This idea can be expressed as an equation:

$$\text{universe} = \text{system} + \text{surroundings}$$

Notice that a system is something that we define. One person might define the system in **Figure 3.2** as the person jumping and the bungee cord. Someone else might define it as the person, the bungee cord, and the bridge that the cord is attached to. We define a system to help us study the system itself, as well as the parts of the surroundings that interact with it.

Energy produces change in a system. In the case of the person and the cord, the system is moving from a greater height to a lesser height. Energy may be added to the system from its surroundings or released from the system to its surroundings. For example, energy would be added to the system from its surroundings if wind pushed the person and the cord off the bridge. Similarly, energy is being released from the system to its surroundings as air resistance provides friction that slows the jumper down.



Figure 3.2 This person and cord above the Nanaimo River on Vancouver Island can be considered a system.
Inferring: What are the surroundings of this system?

Activity

Dropper Popper Dilemma

A dropper popper is a special kind of half-ball. You will invert and release it from head height, waist height, and knee height.

1. Write a hypothesis to predict what you think will happen for each drop.
2. Test your hypothesis. How did it compare with your observations?
3. For each drop, identify the system you observed and its surroundings.
4. Discuss your observations and the following questions with your partner, and then with the class.
 - a) What change(s) did you observe in the system?
 - b) How was energy exchanged between the system and its surroundings?
 - c) What other information about energy and its properties could help you explain your observations?



Before you leave this page . . .

- | | |
|---|--|
| <ol style="list-style-type: none"> 1. a) Why can it be a challenge to observe energy directly? b) How can this challenge be overcome? | <ol style="list-style-type: none"> 2. a) Describe a system that could be applied to your classroom. b) What makes up the surroundings of the system you defined? |
|---|--|

There are different forms of energy.

kinetic energy the energy of motion

potential energy the stored energy of an object as a result of its condition or its position

At the simplest level, energy may be classified into two main types: kinetic energy and potential energy. **Kinetic energy** is the energy of motion. Anything that is moving has kinetic energy. **Potential energy** is the stored energy an object has as a result of its condition or position. For example, the energy stored in the bonds of a chemical compound is a type of potential energy. So is the energy objects have due to their location relative to a reference point, such as the ground. The mountain bike and its rider in **Figure 3.3** have kinetic energy because they are moving and potential energy because they are above the ground. The examples on the next page show different forms of kinetic energy. Examples of the forms of potential energy appear on the pages that follow.



Figure 3.3 This mountain bike and its rider have both kinetic energy due to their motion and potential energy due to their position. **Inferring:** When will the mountain bike and its rider have the most kinetic energy? When will they have the most potential energy?

Kinetic Energy



Mechanical Kinetic Energy

This is the energy of motion of objects that are larger than atoms and molecules. Any object that is moving has mechanical kinetic energy, from the smallest bacteria to the largest galaxies.



Radiant Energy

Radiant energy is the energy of electromagnetic waves that travel or "radiate" from an energy source. For example, light bulbs radiate ultraviolet radiation, visible light, and infrared radiation, which are transformed into thermal energy when they are absorbed by matter. The Sun radiates the entire electromagnetic spectrum. The energy of these waves is often called solar energy. Visible light is often called light energy.

Thermal Energy

This is the energy of the random motion of the particles that make up a substance. Particles of matter are always moving. However, the particles of warmer objects are moving faster than those of cooler objects. In common language, we use the word heat to mean the same thing as thermal energy. In science, however, heat and thermal energy are different. *Heat* is defined as thermal energy that is transferred from one object to another.



Sound Energy

Sound is the energy of vibrations or disturbances of the particles that make up matter. It travels through substances as a pressure wave. As the wave passes through a substance, its particles vibrate back and forth, colliding with nearby particles. In this way, sound energy travels away from its source.

Electrical Kinetic Energy

This is the energy of electrons moving along a wire or other conductor. A load (any electrical appliance) changes the electrical kinetic energy into another form, such as radiant energy. Lightning is also a form of electrical kinetic energy, where the air acts as the conductor.

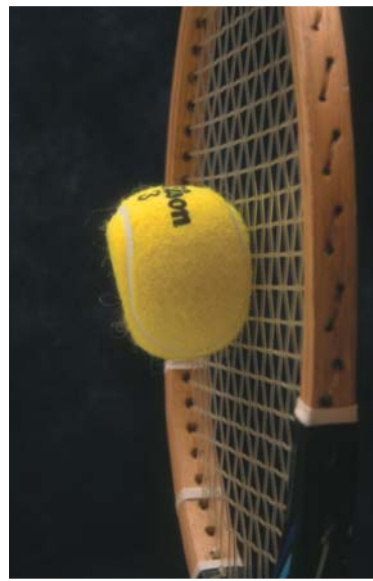


Potential Energy



Chemical Potential Energy

This energy is stored in chemical bonds. Much of human society relies on the chemical potential energy stored in fossil fuels. Some animals, like the firefly shown here, transform chemical potential energy to produce light.



Elastic Potential Energy

This energy is stored in a stretched or compressed object. Elastic potential energy does not just apply to an elastic band or a spring. It applies to any object, like the soles of your shoes when you walk, or the tennis ball and racquet shown here.

Gravitational Potential Energy

This energy is due to the position of an object relative to a reference point, such as the ground. A roller coaster at the top of a large hill has more gravitational potential energy than it does at the bottom. This change results in a hair-raising ride.





Nuclear Energy

This energy is stored within the nucleus of an atom. Nuclear processes can release an enormous amount of energy. Topic 3.2 explores nuclear energy further.



Electrical Potential Energy

This energy is stored by a separation of positive and negative charges, as it is in a cell or battery.



Magnetic Potential Energy

A compass needle moves because it's magnetic and is attracted by Earth's magnetic field. If you prevent the needle from moving, it has magnetic potential energy, as it now has the potential to move.

Activity

Energy Stations

Visit the different energy stations set up around the room, as per your teacher's instructions. At each station, follow the instructions provided. Then identify the type(s) of energy demonstrated at each station.



Before you leave this page . . .

1. Use a Venn diagram to compare kinetic and potential energy.
2. Give one example of each of the following:
 - a) a form of kinetic energy
 - b) a form of potential energy
 - c) a form of energy that has both kinetic and potential energy

CONCEPT 3

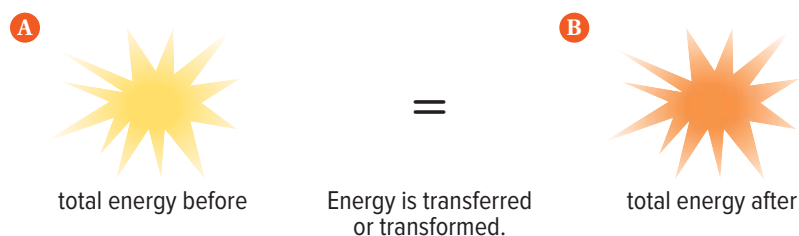
Energy can be transferred or transformed.

law of conservation of energy law stating that energy is neither created nor destroyed, but is transformed from one form of energy to another or transferred from one object to another

Figure 3.4 According to the law of conservation of energy, energy present before energy transfer or transformation

A is equal to energy present afterward **B**. The form of energy may change (the shapes differ in colour), but the amount of energy remains equal (the size of the shape is the same).

Scientists have conducted thousands of experiments to investigate the properties of energy. The results of these experiments are consistent. The total amount of energy present before energy is transferred or transformed is always exactly equal to the total amount of energy present afterwards. In other words, energy is neither created nor destroyed. Instead, it is transformed from one form of energy to another, or transferred from one object to another. This concept is called the **law of conservation of energy** (Figure 3.4).



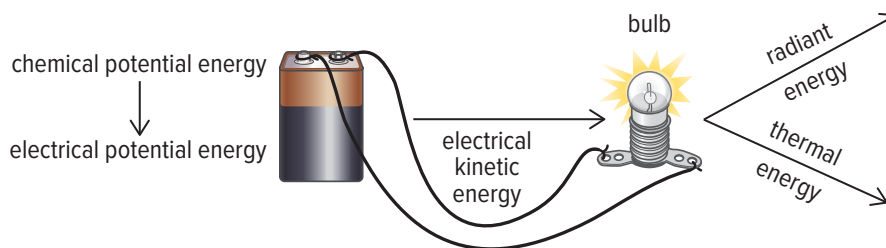
The transfer and transformation of energy often enables useful tasks to be carried out. For example, observe the system in Figure 3.5. Chemical potential energy is transformed into electrical potential energy in the battery. When a light bulb is placed in a closed circuit, electrical potential energy is transformed into electrical kinetic energy, and current flows. As the current flows through the light bulb, the energy is transformed into radiant energy—visible light, with some infrared and ultraviolet radiation—and thermal energy. Because the light energy lets you see, a useful task is carried out.

Energy Transformation, Energy Transfer, and Systems

No energy transformation is 100 percent efficient. Each time that energy changes form, some of it becomes unusable. For example, the system in Figure 3.5 is designed to transform chemical potential energy into light energy. The thermal energy is an unusable byproduct of this energy transformation. In fact, all energy transformations result in some amount of unusable energy. This idea, and how it is linked to the law of conservation of energy, is shown in Figure 3.6.

Connect to Investigation 3-A on page 216

Figure 3.5 Energy is both transformed and transferred in this image, but it is never destroyed. The result is that a useful task is carried out.



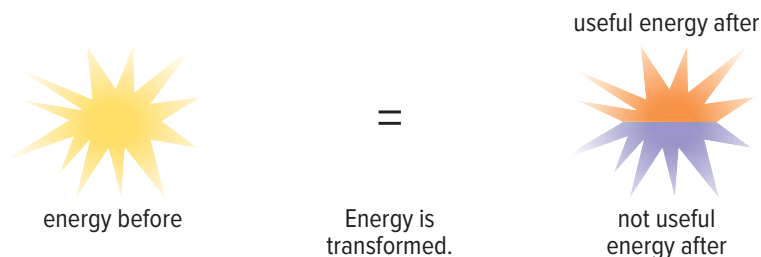


Figure 3.6 When energy is transformed, some of that energy becomes energy that is not useful to carry out a task. However, total energy is still conserved.

Connect to Unit 2
on page 125

Types of Systems

Sometimes, non-useful energy is described as “lost.” However, whether this energy actually leaves the system depends on which type of system it is—open, closed, or isolated. Each type of system is described in terms of the transformation of energy and the transfer of both energy and matter.

- An open system can exchange both energy and matter with its surroundings.
- A closed system can exchange energy, but not matter, with its surroundings.
- An isolated system cannot exchange energy or matter with its surroundings.

Figure 3.7 illustrates the three types of systems.



Open System

An uncovered pot of potatoes boiling on the stove is an open system. Thermal energy is transferred from the stove burner to the pot and its contents, as well as to the surrounding cooler air. As the water boils, thermal energy is also transformed into the mechanical kinetic energy of rising steam. As the steam leaves the pot, the system loses both matter and energy to the surroundings.

Closed System

A pressure cooker with potatoes boiling represents a closed system, because the tightly sealed lid prevents loss of matter and energy to the surroundings in steam. Thermal energy can be transferred into the system from contact between the pot and the stove. It also can be transferred out of the system where the pot contacts the surrounding cooler air and through transformation into radiant energy.



Isolated System

The pot of potatoes inside an insulated container represents an isolated system. In theory, the insulation prevents the exchange of any energy or matter between the system and its surroundings. In reality, energy exchange is significantly reduced, but not eliminated entirely. This is because it is hard to completely isolate a system.

Figure 3.7 An open system, a closed system, and an isolated system are shown.
Applying: Why is it impossible to cook potatoes in an isolated system in real life?

Connect to Investigation
3-B on page 217

Comparing Energy Transfer and Transformation

Whenever a system releases energy, the surroundings absorb it. In the same way, when a system absorbs energy, the surroundings release it. These processes can involve either energy transfer, energy transformation, or both. When energy is *transferred*, it stays in the same form. However, when energy is *transformed*, the form of energy always changes. For example, when one pool ball strikes another in **Figure 3.8**, mechanical kinetic energy is transferred from one ball to the other, but it is also transformed into sound and thermal energy, which are absorbed by the surroundings. We cannot feel the resulting small change in the thermal energy of the surroundings. But, we do hear the vibrating air molecules as sound when it reaches our ears.

Figure 3.8 In pool, mechanical kinetic energy is transferred from one ball to another, but it is also transformed into sound and thermal energy.



Activity

Modelling Energy Transfer and Transformation

1. Build a catapult with the materials provided by your teacher.
2. Experiment with the design of your catapult to determine changes that affect its performance.
3. Record all the ways that energy is transformed and transferred as your catapult operates.
4. Using your observations, try to determine what variables affected energy transformation

in the catapult. To answer this question, consider how changes to your design affected a certain function. For instance, if you used an elastic band in your design, did the length or thickness of the band affect the distance the catapult was able to throw an object? What type of energy might have been affected by this variable?



Before you leave this page . . .

1. Describe the law of conservation of energy.
2. How do energy transfer and transformation differ? How are they similar?
3. Use an example from your everyday life to show how you could change an open system to
a) a closed system and **b)** an isolated system.

Physical quantities contribute to different forms of energy.

Activity

Energy Thought Experiments

Thought experiments are experiments done in your mind. Complete the ones below to determine the physical quantities that contribute to mechanical kinetic energy and gravitational potential energy. Justify your answers.

Experiment 1: Mechanical Kinetic Energy

1. A bowling ball and a table tennis ball are rolling toward your foot with the same speed. Which would you try hardest to avoid?
2. Two identical golf balls are rolling toward your foot. One is coming at you slowly, and the other is coming toward you very quickly. Which would you try hardest to avoid?
3. Based on your answers to questions 1 and 2, what physical quantities do you think affect mechanical kinetic energy?

Experiment 2: Gravitational Potential Energy

1. A bowling ball and a tennis ball are dropped from a height of 10 m. Which one would hit the ground harder?
2. If one bowling ball is dropped from a height of 10 cm and another is dropped from a height of 10 m, which one would hit the ground harder?
3. Which golf ball would hit the surface with the greatest impact: a ball dropped from 1 m above Earth's surface or a ball dropped from 1 m above the Moon's surface? (Refer to Appendix C.)
4. What physical quantities do you think affect gravitational potential energy?

Experience tells you that different physical quantities contribute to different forms of energy. For example, you know that if you throw different balls at a target, as in **Figure 3.9**, the mass and speed of each ball affect whether it can knock over the target. This means that both mass and speed contribute to the mechanical kinetic energy of a system.

On the other hand, observing the impact of falling objects, as in **Figure 3.10**, tells you that mass and the height from which they begin their fall contribute to an object's gravitational potential energy. You also know that Earth exerts a gravitational pull on objects. Any object that is thrown up comes down again. This helps you understand that the force of gravity affects gravitational potential energy as well.

Figure 3.9 A faster ball with greater mass would be more likely to knock over the target in this game.



Figure 3.10 Gravity, the mass of these rocks, and the height from which they begin their fall contribute to their gravitational potential energy.

Visualizing Physical Quantities That Affect Kinetic and Potential Energy

Figure 3.11 shows how different physical quantities affect the mechanical kinetic and gravitational potential energy of a person and a sled. In the figure, gravitational potential energy (GPE) is being transformed into mechanical kinetic energy (KE) and back again as the sled travels down one hill and up another. Note that the gravitational potential energy of an object is always compared to a reference position. This can be any position you choose, but it is often the surface toward which the object is falling or might fall. In **Figure 3.11**, the reference position is the ground at the base of each hill.



At the starting point, the sled is sitting on the highest hill.

- The sled is not moving, so it has no KE.
- The sled is at the greatest height, so it has maximum GPE.



The sled is midway down the highest hill.

- The sled is increasing in speed. KE is increasing.
- Height above the reference point is decreasing, so GPE is decreasing.

Figure 3.11

Gravitational potential energy (GPE) is transformed into mechanical kinetic energy (KE) and back again in this sled run. After the sled stops on the second hill, it will slide backward down the hill as the remaining GPE is once again transformed into mechanical KE. **Analyzing: Which physical quantity is not changing in this scenario for KE? for GPE?**



The sled is at the bottom of the highest hill.

- The sled is at its greatest speed. It has maximum KE.
- The sled is at the reference point, so it has no GPE.

Why doesn't the sled make it up the second hill?

Gravitational potential energy isn't just transformed into mechanical kinetic energy as the sled travels. It's also transformed into unusable thermal energy and sound energy. This unusable energy is the product of *friction*. When two surfaces touch, attractive forces form between the atoms that are in contact with each other. When the sled travels over the snow, these forces are continually broken and reformed, producing thermal energy and sound energy. The farther the sled travels, the more energy leaves the system (person and sled) in this way. When it reaches the second hill, it doesn't have enough energy to go all the way up.



The sled stops when it is almost at the top of the second hill. It does not have enough mechanical kinetic energy to keep going.

- The sled is not moving, so it has no KE.
- Height above the reference point is less than it was at the top of the highest hill, so GPE is less than it was at the beginning of the sled run.

The sled is partway up the second hill.

- The sled is decreasing in speed. KE is decreasing.
- Height above the reference point is increasing, so GPE is increasing.

Connect to Investigation 3-C on page 218

Energy Equations

Many forms of energy can be described mathematically. The equations describing mechanical kinetic energy and gravitational potential energy are given below. Your teacher may choose to assign the two activities that follow, which introduce you to approaches for how to solve problems that involve these forms of energy.

Mechanical Kinetic Energy (KE)

The formula shown here is the mathematical relationship between an object's mechanical kinetic energy and its mass and velocity. The term *velocity* refers to the speed that something is moving in a specific direction.

$$E_k = \frac{1}{2}mv^2$$

Quantity	Symbol	SI unit
mechanical kinetic energy	E_k	J (joule)
mass	m	kg (kilogram)
velocity*	v	$\frac{m}{s}$ (metres per second)

* Velocity is squared in the calculations for mechanical kinetic energy.

Gravitational Potential Energy (GPE)

The formula shown here is the mathematical relationship between an object's gravitational potential energy and its mass, the acceleration due to gravity (9.8 m/s^2), and the change in height. Acceleration due to gravity is the rate at which an object's velocity increases when it falls toward Earth and there is no friction to slow it down.

$$E_g = mg\Delta h$$

Quantity	Symbol	SI unit
gravitational potential energy	E_g	J (joule)
mass	m	kg (kilogram)
acceleration due to gravity	g	$\frac{m}{s^2}$ (metres per second squared)
change in height (from reference position)	Δh	m (metre)

Note: The SI unit for energy is the joule (J), which is equivalent to a $\text{kg} \cdot \frac{\text{m}^2}{\text{s}^2}$. One joule is needed to lift one large kiwi fruit (about 100 g) a distance of 1 m. Since the energy represented by a joule is very small, it is often expressed in kilojoules (1 kJ = 1000 J).

Activity

Energy Equation Thought Problems

Use the equations on the previous page to determine how mechanical kinetic and gravitational potential energy are affected in the following problems. Explain your answer in each case.

1. Two cars are driving at the same velocity, but one has twice the mass of the other. Is the mechanical kinetic energy of the larger car two times, three times, or four times that of the smaller car?
2. You are skateboarding to school and realize that you are late. If you double your pace, by what factor would your mechanical kinetic energy increase?
3. Two rock climbers of the same mass are climbing a cliff.
 - a) One stops to rest at a position that is 50 m above the ground. The other climber stops at a height of 25 m above the ground. Which one has greater gravitational potential energy?
 - b) Would the climbers have more or less gravitational potential energy if they were climbing a cliff on the Moon?

Activity

Energy Equation Calculation Problems

Study the following sample problem to understand how to use the two energy equations you just learned. Then solve the practice problems.

Sample Problem

A 45 kg boulder is sitting on a cliff that is 125 m high.

- a) How much gravitational potential energy does the boulder have with respect to the bottom of the cliff?
- b) If the boulder rolls off the cliff and falls to the ground, at what velocity will it be falling just before it hits the ground? Ignore friction with the air.

Solution

- a) Substitute the known values into the gravitational potential energy equation.

$$\begin{aligned}E_g &= mg\Delta h \\&= 45 \text{ kg} \times 9.8 \frac{\text{m}}{\text{s}^2} \times 125 \text{ m} \\&= 55\,125 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2} \\&\cong 5.5 \times 10^4 \text{ J}\end{aligned}$$

The answer has two significant digits because two is the fewest number of significant digits in the data. The answer has been converted into scientific notation.

- b) Rearrange the mechanical kinetic energy equation to determine the velocity (v). To do this, divide both sides of the equation by m and multiply both sides by 2. Then take the square root of both sides of the equation.

$$E_k = \frac{1}{2}mv^2$$

$$\frac{E_k}{m} = \frac{\frac{1}{2}mv^2}{m}$$

$$\frac{E_k}{m} = \frac{1}{2}v^2$$

$$\frac{E_k}{m} \times 2 = \frac{1}{2}v^2 \times 2$$

$$\frac{E_k}{m} \times 2 = v^2$$

$$v = \sqrt{\frac{E_k \times 2}{m}}$$

Now substitute the known values into the mechanical kinetic energy equation. Because energy is conserved, all the potential energy the boulder had at the top of the cliff is transformed into mechanical kinetic energy at the bottom. Thus, the total mechanical kinetic energy of the boulder at the instant before it hits the ground is 55 125 J, the unrounded answer from part a). When you use one answer in a calculation for another part of a question, always use the unrounded number. Again, the answer has two significant digits because two is the fewest number of significant digits in the data.

$$v = \sqrt{\frac{55\,125 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2} \times 2}{45 \text{ kg}}}$$

$$v = \sqrt{2450 \frac{\text{m}^2}{\text{s}^2}}$$

$$v = 49.497 \frac{\text{m}}{\text{s}}$$

$$v \cong 49 \frac{\text{m}}{\text{s}}$$

Practice Problems

1. A 54 kg skier, including equipment, stands at the top of a black diamond ski run. The vertical distance to the bottom of the run is 420 m. What is the gravitational potential energy of the skier relative to the bottom of the ski run?
2. A satellite has a mass of 689 kg and travels at a speed of 27 000 km/h (7500 m/s). How much mechanical kinetic energy does the satellite have?
3. A bowling ball is rolling down the lane at 2.8 m/s. If it has a mechanical kinetic energy of 25.5 J, what is its mass?
4. A person who has a mass of 65 kg goes on the Sky Tower ride at an amusement park. The ride is simply a free fall from the top of a tower into a net below. If the person reaches a final velocity of 24.6 m/s just before hitting the net, from what height did the rider drop? Ignore friction with the air.



Before you leave this page . . .

1. What physical quantities affect **a)** mechanical kinetic energy and **b)** gravitational potential energy?
2. Why might it appear that the law of conservation of energy does not apply to the sled in **Figure 3.11**?

Check Your Understanding of Topic 3.1

QP Questioning and Predicting
 PC Planning and Conducting
 PA Processing and Analyzing
 E Evaluating
AI Applying and Innovating
C Communicating

Understanding Key Ideas

1. Use the photo to answer the following questions. **PA**



- a) Identify four types of energy in the photo.
 - b) Classify each type of energy as either kinetic or potential. Explain your answer in each case.
2. Explain why thermal energy is a form of kinetic energy. **PA**
 3. a) Identify an energy transformation that occurs in your daily life.
 b) Identify an energy transfer you experience every day.
 c) Explain how the energy transformation you described differs from the energy transfer. **PA AI**
 4. Why is no process of energy transformation 100 percent efficient? **PA**
 5. Draw a sketch and write a caption that relates the properties of an open mug, a travel mug covered by a lid, and a closed, insulated thermos to the three types of systems. Your answer should explain the differences among the three types of systems. **PA C**
 6. Imagine that a referee tosses a basketball straight up and it falls to the floor.
 - a) At what point in its journey would the ball's gravitational potential energy be at a maximum?

- b) At what point would the ball have no gravitational potential energy?
- c) At what point would the ball's mechanical kinetic energy be at a maximum?
- d) At what points would the ball have no mechanical kinetic energy? (Hint: This occurs at more than one point.) **QP PA AI**

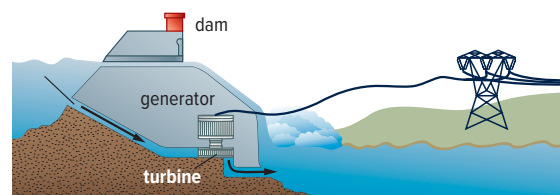
Connecting Ideas

7. a) Identify a system and surroundings in the photo.
 b) Describe how energy enters the system.
 c) Draw a diagram to show how energy is being transformed within the system.
 d) Describe how energy leaves the system.
 e) Explain how the law of conservation of energy applies to the system. **PA AI C**



Making New Connections

8. Explain why the amount of electrical energy generated by a hydro dam depends on both the mechanical kinetic energy and the gravitational potential energy of the water shown in the diagram. **PA AI**



Skills and Strategies

- Questioning and Predicting
- Planning and Conducting
- Processing and Analyzing
- Evaluating
- Communicating

What You Need

- materials your team's design requires

Build a Rube Goldberg Machine

Rube Goldberg was a prize-winning cartoonist famous for his drawings of fun machines that perform simple tasks in the most complicated way imaginable. Rube Goldberg machines usually employ a lot of different energy transfers and transformations to get the job done.



Question

How can you design and build a machine that uses many different energy transformations and transfers to perform a simple task?

Procedure

1. With your group, choose a task you want to carry out.
2. Design your machine. Sketch your design, complete a list of materials that you will need, and record any safety precautions.
3. Have your teacher approve your sketch, materials list, and safety precautions. Collect the materials you need and build your machine.
4. Test your machine, making any required modifications to improve its performance.

Process and Analyze

1. Identify each different energy transfer and/or transformation that occurs when you run your machine.

Evaluate and Communicate

2. a) Did your machine perform its task as planned? Explain.
b) How might you change your design to improve its performance?

Skills and Strategies

- Questioning and Predicting
- Planning and Conducting
- Processing and Analyzing
- Evaluating
- Communicating

What You Need**Suggested Materials****Question A**

- sand
- empty plastic containers with lids
- laboratory balance
- measuring tape
- scoop
- tray

Question B

- toy car track
- marbles of different sizes
- measuring tape
- clamps and supports

Question C

- water
- ice
- thermometers
- plastic containers or cups
- variety of insulative materials
- tape

Investigating Energy Transformation and Transfer

In this investigation, you will design and carry out a controlled experiment to answer a question about energy transformation and/or transfer.

Questions

- Does the mechanical kinetic energy of a falling object increase if its mass increases?
- When an object rolls down a U-shaped ramp and back up the other side, can it reach the height from which it was released?
- Can you construct an isolated system?
- What question of your own can you investigate?

Procedure

- Choose one of the above questions, and develop a hypothesis for it.
- Design a procedure to test your hypothesis. Decide on the independent and dependent variables. Determine how you will keep other factors constant.
- Create a data table to record your observations. Plan how you will analyze your data to determine if they support your hypothesis.
- Have your teacher approve your procedure. Then carry it out.
- Summarize your results using a method assigned by your teacher.

Process and Analyze

- How well do your data support your hypothesis?
- Describe the energy transformation or transfer in your experiment.

Evaluate and Communicate

- Exchange procedures with another group that answered the same question. After reading their procedure, suggest one way that you might improve your own.

Skills and Strategies

- Questioning and Predicting
- Planning and Conducting
- Processing and Analyzing
- Evaluating
- Communicating

Safety



- Use caution if handling scissors or other sharp utensils.

What You Need

- materials your team's design requires

Roller Coaster Physics

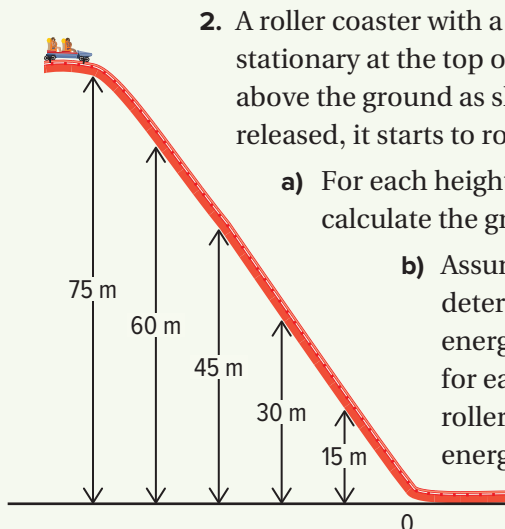
PART A: Roller Coaster Simulation

Question

How are the gravitational potential energy and the mechanical kinetic energy of a roller coaster related?

Procedure

1. Make a table to record answers to the Step 2 questions. Include columns for height (m), total energy (J), gravitational potential energy (J), mechanical kinetic energy (J), and velocity (m/s).

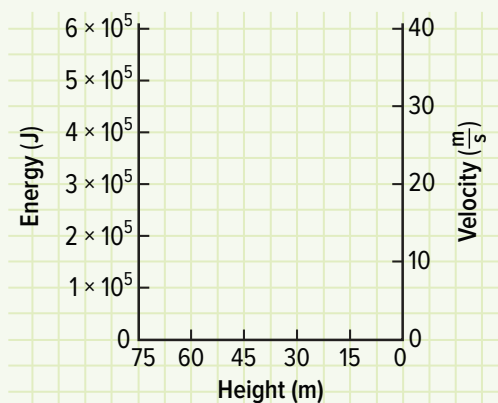


2. A roller coaster with a mass of 800 kg sits stationary at the top of a section of track, 75 m above the ground as shown. When the brake is released, it starts to roll down the track.

a) For each height indicated in the diagram, calculate the gravitational potential energy.

b) Assuming that there is no friction, determine the mechanical kinetic energy and calculate the velocity for each of the heights of the roller coaster. (Hint: Recall that energy is conserved.)

- c) Use your answers to graph how gravitational potential energy, mechanical kinetic energy, and velocity change as the roller coaster changes height. Use different colours for the graphs.



Process and Analyze

1. a) Describe the relationship between the gravitational potential energy and the mechanical kinetic energy of the roller coaster on your graph.
b) Describe the shape of the three lines in the graph. Explain why the shape is different.
2. At what point does the roller coaster have a maximum value of the following? Justify your responses.
 - a) gravitational potential energy
 - b) mechanical kinetic energy
 - c) velocity
3. In your calculations, you assumed that the roller coaster was frictionless. All real roller coasters encounter friction. Describe how the actual values of the variables would differ, or not differ, from your calculated values for a real roller coaster.

PART B: Designing and Testing a Model Roller Coaster

You are part of a team that is working to design the world's fastest roller coaster, *The Vortex*. Your responsibility on the team is to design, build, and test the initial model for the roller coaster.

Question

How can you increase the velocity of a roller coaster?

Procedure

Engineers who design roller coasters and other vehicles build scale models to test their ideas. With a partner or group, you will design, build, and test a model roller coaster.

1. With your partner or group, discuss the following questions.
 - What physical quantities would affect the velocity of a roller coaster?

- What will you use to model the roller coaster?
 - What will you use to make a track?
 - How will you be sure that your roller coaster will stay on its track?
 - How will you support your track?
 - How will you ensure that your roller coaster will have enough energy to reach the end of the track?
 - How could you reduce friction experienced by the roller coaster?
2. Design your roller coaster model. Sketch your design, complete a list of materials that you will need, and record any safety precautions.
 3. Have your teacher approve your sketch, materials list, and safety precautions. Collect the materials you need and build your model.
 4. Test your model, making any required modifications to improve its performance.

Process and Analyze

1. a) Calculate the gravitational potential energy of your roller coaster at each high point on your track.
b) Determine the mechanical kinetic energy and velocity of the roller coaster at points where you expect it to be moving at near maximum velocity. Assume no friction.

Evaluate and Communicate

2. Compare your model with others in the class. Based on your observations, decide if and how you could improve your model. Modify and retest your model if time allows.
3. How would you promote your roller coaster design based on scientific evidence? Use a medium of your choice to create an evidence-based promotion of your rollercoaster.